

Git-Vista iPad Design

Status: proposed

Git-Vista should not be a desktop Git UI made larger. Its primary interface is a touch surface that may also have an Apple Pencil, trackpad, keyboard, split-screen constraint, or external display. Every essential workflow must remain complete with one finger.

Interaction Principles

1. **Touch is complete.** Hover, right-click, and keyboard shortcuts are optional accelerators.
2. **Selection is not execution.** Selecting a commit, range, hunk, or rebase row never mutates a repository.
3. **Plans are tangible.** History-changing operations become editable visual plans followed by a review and explicit execute step.
4. **Context remains visible.** The current repository, worktree, branch, dirty state, and running operation remain visible during navigation.
5. **Density adapts to space, not device names.** Use container queries and input capabilities rather than an iPad user-agent branch.
6. **Precision is optional.** Pencil precision improves range and hunk selection, but no command requires a Pencil.
7. **Recovery is part of the interaction.** A completed destructive operation shows its result and available recovery action in place.

Use a minimum interactive target of 44 by 44 CSS pixels, adequate separation, visible pressed/selected/focus states, and no destructive edge swipe.

Adaptive Shell

Wide Landscape or External Monitor

```
+-----+-----+-----+
| repositories | history / status / diff | inspector / plan | |
| and worktrees | | | |
| | persistent action bar | operation progress |
+-----+-----+-----+
```

- The left rail selects repository and worktree; it is not a file explorer.
- The center canvas owns graph navigation, status, or comparison.
- The inspector preserves selected-object context while the graph remains usable.
- A hardware keyboard may expose a command palette and shortcuts.

Portrait

```
+-----+
| repository / worktree / branch / sync state |
+-----+
|                                             |
| primary graph, status, or diff surface      |
|                                             |
+-----+
| contextual action dock                     |
+-----+
| inspector or operation plan as bottom sheet |
+-----+
```

The bottom sheet has detents for summary, half-height, and full-height content. It must not cover the selected graph item without an obvious way to restore context.

Split Screen and Narrow Stage Manager Windows

- Collapse repository navigation into a modal sheet.
- Replace persistent inspector with a full-height sheet.
- Present one primary task at a time: graph, status, diff, plan, or result.
- Keep branch and dirty-state indicators in the compact header.
- Preserve selected commit, viewport, and unfinished form when the width changes.

External Monitor

- Increase information density; do not simply scale controls and whitespace.
- Permit graph plus side-by-side diff plus inspector when width allows.
- Maintain touch-sized actions because the external display may still be touch adjacent or controlled from the iPad.
- Never move a critical confirmation to a different display unexpectedly.

Input Model

Use Pointer Events as the unified input layer. Inspect `pointerType`, pressure, tilt, and contact geometry only to enhance an interaction. Preserve a click, touch, and keyboard path for the same command.

Input	Primary behavior
One-finger tap	Select; a second explicit control opens or executes
One-finger drag	Pan graph or scroll a list; never reorder unless a handle is held
Pinch	Zoom graph around the gesture centroid
Long press	Open contextual actions with the selected object identified
Two-finger tap	Optional back/cancel accelerator; never the only path
Pencil tap	Precise selection of commit, connector, line, or hunk
Pencil drag	Select a range, annotate a teaching diagram, or reorder by handle
Trackpad/mouse	Hover previews and contextual menu as enhancements
Keyboard	Command palette, navigation, range selection, and shortcuts

Pressure must never confirm, delete, push, or rewrite history. Pencil hover is not available on all devices and cannot expose required information exclusively.

Core Workflows

Graph Navigation

- Tap a commit to select it and show a summary sheet.
- Expand the sheet for metadata, refs, changed files, and operation entry points.
- Pinch zoom changes semantic detail, not only geometric scale.
- A visible minimap or position rail supports very large histories.
- Branch/ref filters are chips in compact mode and a panel in wide mode.
- Range selection uses an explicit "Select range" mode with clear endpoints.

Status and Partial Staging

Desktop checkbox lists are insufficient for touch. Use file cards with status, summary, and an explicit stage control. Opening a file shows a virtualized diff:

- Tap a hunk header to select the hunk.
- Use Pencil or a range handle to refine to lines where Git permits it.
- Keep selected additions/deletions visually distinct from diff syntax colors.
- Show the generated patch preview before applying a partial stage/discard.
- Make discard harder than unstage and state exactly which content will be lost.

Commit

- Keep staged scope visible beside or above the commit form.
- Preserve draft messages across suspension and layout changes locally, but do not sync them to other devices by default.
- Surface hooks and signing as operation progress rather than a frozen button.
- Make amend a distinct mode with the affected commit and force-push consequence visible before execution.

Interactive Rebase

Represent rebase as a board of ordered commit cards. Each row has a large reorder handle and an action selector for pick, reword, edit, squash, fixup, or drop. Dragging only edits the plan. A review screen shows:

- Original and proposed topology.
- Branches and remotes affected.
- Likely conflicts and dirty-worktree blockers.
- The checkpoint/recovery reference.
- Whether a later force push may be required.

Conflict Resolution

Avoid three tiny side-by-side editors in portrait. Present one conflict at a time with base, ours, theirs, and result as selectable views. Provide whole-block choices, then optional line-level editing. The continue/abort bar remains visible and identifies the active merge, rebase, cherry-pick, or stash operation.

Worktrees and Stash

Treat worktrees as first-class workspaces in the repository rail. A "switch task" flow should recommend a worktree, commit, or stash based on the user's intent and explain the tradeoff rather than hiding everything behind a generic stash action.

Feedback, Motion, and Failure

- Use motion to preserve spatial context when opening an inspector or changing graph scale. Respect `prefers-reduced-motion`.
- Do not rely on platform haptics; browsers cannot provide consistent haptic feedback.
- Show optimistic selection but never optimistically report a Git mutation as complete.
- Running operations survive navigation. Progress belongs to a global operation center as well as the initiating screen.
- On reconnect, reconcile by operation ID and repository generation rather than replaying the last request.
- Restore viewport and selection after Safari suspends or reloads the PWA.

Accessibility

- Expose the commit graph as both a visual canvas and a navigable semantic list.
- Describe parent relationships, refs, and current branch in accessible text.
- Provide non-color indicators for lanes, selection, staged state, and conflicts.
- Support Dynamic Type-like browser text scaling without clipping actions.
- Make every gesture workflow available through visible controls and keyboard.
- Announce operation start, progress changes, completion, and recoverable failure.
- Test VoiceOver with touch exploration; pointer support alone is not access.

Browser and PWA Requirements

- Use safe-area insets and modern dynamic viewport units for installed mode.
- Handle `visibilitychange`, network loss, process restart, and stale service worker assets explicitly.
- Pin protocol compatibility between the cached frontend and running server.
- Keep app-shell caching separate from private repository data caching.
- Virtualize graph rows and diffs, bound decoded payloads, and avoid retaining entire repository histories in WASM memory.
- Test current Safari/iPadOS plus Chromium and Firefox desktop; do not depend on a WebKit-only feature for correctness.

Validation Matrix

Every release candidate should cover:

- 11-inch iPad portrait and landscape with finger only.
- iPad split screen at narrow and medium widths.
- Stage Manager resize while a form and graph selection are active.
- Apple Pencil selection plus finger panning in the same session.
- Magic Keyboard and trackpad.
- External display at desktop width.
- VoiceOver, text zoom, reduced motion, and increased contrast.
- 200 ms latency through an SSH tunnel and a mid-operation disconnect.
- A large repository with long history, large diffs, and binary files.
- Offline (browser reports no network, or the tunnel dies mid-session): the browser's own network error surfaces (`network_error_text`), 22a's write guard refuses mutations before they hit the wire, and 22b's banner and disabled controls appear. Per ADR 0032, this is a test that the app **fails loudly** — not a test of a cached or offline-readable view, which the ADR rejects outright.

Note on #75's cache criteria. #75's original acceptance criteria "Private diffs are not cached by default" and "Cache clear and export controls exist" are satisfied vacuously here: the frontend has no client-side cache of `/api` data to clear or export in the first place. Verified against `crates/git-vista/src` (2026-07-31): the only client-side storage is two `localStorage` UI-preference booleans in `prefs.rs` (icon style, per-node icons); `grep -rn "IndexedDb\|caches\.\|Blob\|localStorage" crates/git-vista/src` turns up nothing else. ADR 0032 forbids adding a cache of `/api` data, so this is expected to stay true, not a gap to close.

This is the narrower, ADR-consistent reading: ADR 0032 rules out a service worker and any "make offline look normal" outcome, but it does not unambiguously rule out every possible non-service-worker, application-level store. Whoever closes #75 (22d) should treat this as an explicit invitation to override — if the original intent behind those two criteria was such a store, say so and reopen the question rather than letting this reading stand by default.

Reference

Pointer Events provide the cross-device pointer model, including touch, pen, mouse, pressure, and tilt: <https://www.w3.org/TR/pointerevents3/>.